



ALLIANCE
UNIVERSITY
Private University established in Karnataka State by Act No. 34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi.

NAAC
GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

Title: Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

Submitted By:

Lakhan Singh

Reg. No.: 2411021240058

Date: 28/08/2026

Submitted To:

Faculty Name: Mr. Aman Kumar Sharma

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru – 562106

Experiment No.: 2

Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

GitHub Link : <https://github.com/Lakhan07AU/Bussiness-Analytics-Projects/tree/main/Experiment-2>

Problem Statement

Management wants to quickly understand sales performance across products, categories, and regions. Create appropriate Power BI visualizations to identify key business patterns.

Note: Specifically use a the following Business dataset

<https://www.kaggle.com/datasets/naveenkumar20bps1137/sample-superstore>

Step 1: Data Collection

The Sample Superstore dataset was collected in CSV format and used as the primary data source for the analysis.

The dataset contains 9,994 sales records and includes the following important fields:

- Ship Mode
- Segment
- Country
- City
- State
- Postal Code
- Region
- Category
- Sub-Category
- Sales
- Quantity
- Discount
- Profit

The dataset provides sufficient information to analyze sales performance across different products, categories, regions, and states.

Step 2: Data Import into Power BI

The following steps were performed to import the dataset:

1. Opened Power BI Desktop.
2. Selected Get Data.
3. Selected the CSV file containing the Sample Superstore dataset.
4. Loaded the dataset into Power BI.
5. Verified that all required columns were imported correctly.
6. Checked the data types of numerical and categorical fields.

The imported table was used as the primary data source for creating the dashboard.

Step 3: Data Cleaning and Transformation

The data was checked and prepared before creating the visualizations.

The following operations were performed:

- Checked column names and data types.
- Verified numerical fields such as Sales, Profit, Quantity, and Discount.
- Verified categorical fields such as Category, Sub-Category, Region, and State.
- Checked for missing or inconsistent values.
- Ensured that Sales, Profit, and Quantity were available as numerical fields.
- Set State as a geographical field for the map visualization.
- Prepared the dataset for analysis and visualization.

This ensured that the data could be correctly aggregated and represented in Power BI.

Step 4: Data Modeling

Since the analysis was performed using the Sample Superstore dataset, the project primarily uses a single dataset/table.

The following fields were used for analysis:

Field	Purpose
Category	Category-wise sales analysis
Sub-Category	Product-level sales analysis
Region	Regional sales comparison
State	Geographical sales analysis
Sales	Primary sales metric
Profit	Profitability analysis
Quantity	Quantity sold
Discount	Discount analysis
Segment	Customer segment analysis

The geographical State field was also used in the map visualization to identify the distribution of sales across the United States.

Step 5: DAX Measures

DAX measures were created to calculate the major business performance indicators.

1. Total Sales

Total Sales = SUM('SampleSuperstore'[Sales])

2. Total Profit

Total Profit = SUM('SampleSuperstore'[Profit])

3. Total Quantity

Total Quantity = SUM('SampleSuperstore'[Quantity])

4. Profit Margin

Profit Margin =

DIVIDE(
 [Total Profit],
 [Total Sales],
 0
)

The Profit Margin represents the proportion of sales retained as profit.

Formula

Profit Margin = Total Profit / Total Sales × 100

The overall profit margin obtained from the dataset is approximately 12.47%.

Step 6: Dashboard Development

A Sales Performance Dashboard was developed in Power BI.

The dashboard contains the following major components:

KPI Cards

Four KPI cards were created:

1. Total Sales
2. Profit
3. Quantity
4. Profit Margin

The dashboard displays approximately:

- Total Sales: \$2.30M
- Total Profit: \$286.39K
- Total Quantity: 38K
- Profit Margin: 12.47%

These KPIs provide management with a quick overview of overall business performance.

Step 7: Sales by Category

A column chart was created to compare sales across product categories.

The categories are:

- Technology
- Furniture
- Office Supplies

Sales Performance

Category	Sales
Technology	\$836.15K
Furniture	\$742.00K
Office Supplies	\$719.05K

Insight:

Technology is the highest-performing category with approximately \$836.15K in sales. Furniture follows with approximately \$742.00K, while Office Supplies contributes approximately \$719.05K. Therefore, Technology represents the strongest category in terms of sales.

Step 8: Sales by Region

A horizontal bar chart was created to compare sales across four regions.

Region	Sales
West	\$725.46K
East	\$678.78K
Central	\$501.24K
South	\$391.72K

Insight:

The West region has the highest sales at approximately \$725.46K, followed by the East region at approximately \$678.78K.

The South region has the lowest sales at approximately \$391.72K.

This indicates a significant difference in sales performance between the strongest and weakest regions.

Step 9: Sales by Sub-Category

A horizontal bar chart was created to identify the highest-selling product sub-categories.

The top 10 sub-categories are:

Rank	Sub-Category	Sales
1	Phones	\$330.01K
2	Chairs	\$328.45K
3	Storage	\$223.84K
4	Tables	\$206.96K
5	Binders	\$203.41K
6	Machines	\$189.24K
7	Accessories	\$167.38K
8	Copiers	\$149.53K
9	Bookcases	\$114.88K
10	Appliances	\$107.53K

Insight:

Phones generate the highest sales among the displayed sub-categories, closely followed by Chairs. Storage, Tables, and Binders are also significant contributors to overall sales.

Step 10: Geographical Sales Analysis

A map visualization was created using State and Sales.

Configuration

- Location: State
- Legend/Size: Sales

The map displays the geographical distribution of sales across different states in the United States. The visualization helps management identify areas with comparatively high and low sales.

Top States by Sales

Rank	State	Sales
1	California	\$457.69K
2	New York	\$310.88K
3	Texas	\$170.19K
4	Washington	\$138.64K
5	Pennsylvania	\$116.51K

Insight:

California is the strongest state in terms of sales, generating approximately \$457.69K. New York is the second-highest contributor with approximately \$310.88K.

The geographical visualization makes it easier to identify important sales markets and regional opportunities.

Step 11: Dashboard Formatting

The dashboard was formatted to provide a clear and professional presentation.

The following formatting elements were applied:

- Added the title Sales Performance.
- Created four KPI cards.
- Used a consistent purple-based visual theme.
- Arranged charts in a structured layout.
- Used column charts for category comparison.
- Used horizontal bar charts for regional and sub-category comparison.
- Added a geographical map for state-level sales analysis.
- Adjusted chart sizes and positions.
- Added appropriate chart titles.
- Formatted numerical values using compact notation such as M and K.
- Ensured that the dashboard was easy to understand at a glance.

The final dashboard provides a centralized view of sales performance.

Step 12: Analysis and Business Insights

The dashboard provides several important business insights.

1. Overall Sales Performance

The business generated approximately \$2.30 million in total sales.

2. Profitability

The total profit is approximately \$286.39K, resulting in a profit margin of approximately 12.47%.

3. Category Performance

Technology is the highest-performing category, generating approximately \$836.15K in sales.

4. Regional Performance

The West region has the highest sales, while the South region has the lowest sales.

5. Product Performance

Phones and Chairs are the two highest-selling sub-categories in the dashboard.

6. Geographical Performance

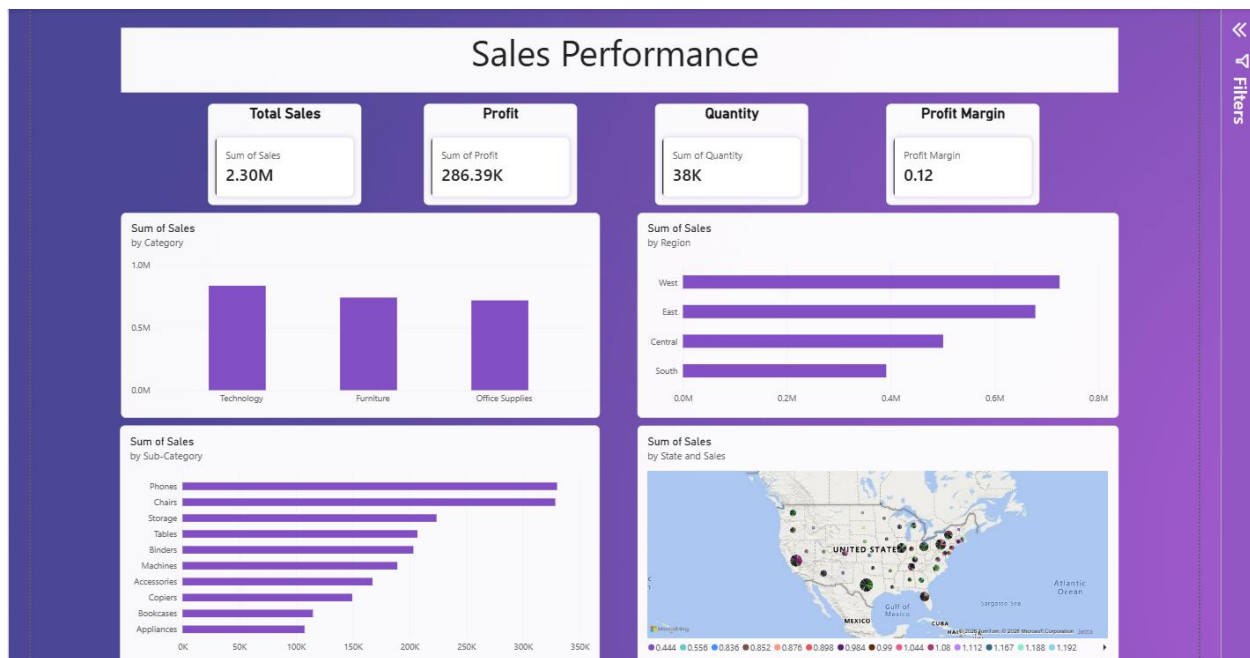
California is the highest-selling state, followed by New York.

7. Management Implication

Management can use these insights to:

- Focus on high-performing categories.
- Identify successful products and sub-categories.
- Investigate the reasons for lower sales in the South region.
- Strengthen sales strategies in high-performing states.

- Use geographical information to identify potential growth markets.
- Monitor profitability along with sales rather than relying only on revenue.



Step 13: Final Dashboard

After completing data import, cleaning, DAX calculations, visualization, and dashboard formatting, the final Sales Performance Dashboard was developed.

The dashboard contains:

- Total Sales KPI
- Profit KPI
- Quantity KPI
- Profit Margin KPI
- Sales by Category chart
- Sales by Region chart
- Sales by Sub-Category chart
- Sales by State map

The final dashboard provides management with a centralized and visual representation of sales performance across products, categories, regions, and geographical locations.

The dashboard screenshot shows the four KPI cards at the top, followed by category and regional sales charts, a sub-category ranking chart, and a state-level sales map.

Conclusion

The Sales Performance Dashboard was successfully developed using Power BI and the Sample Superstore dataset.

The data was imported, checked, cleaned, and prepared for analysis. DAX measures were created

to calculate important business KPIs such as Total Sales, Total Profit, Total Quantity, and Profit Margin.

Multiple visualizations were created to analyze business performance, including column charts, horizontal bar charts, KPI cards, and a geographical map.

The analysis shows that Technology is the leading category, the West is the strongest region, Phones are the highest-selling sub-category, and California is the leading state in terms of sales.

Overall, the project demonstrates how Power BI can transform raw sales data into an interactive business intelligence dashboard, allowing management to quickly understand business performance and make data-driven decisions.

Learning Outcomes

Through this project, I learned:

1. Data Collection

Learned how to work with a business dataset containing sales, product, geographical, and transactional information.

2. Data Import

Learned how to import a CSV dataset into Power BI Desktop.

3. Data Cleaning and Preparation

Learned how to check data types, missing values, categorical fields, and numerical fields before analysis.

4. DAX

Learned how to create DAX measures for calculating:

- Total Sales
- Total Profit
- Total Quantity
- Profit Margin

5. Data Visualization

Learned how to create:

- KPI cards
- Column charts
- Horizontal bar charts
- Map visualizations

6. Geographical Analysis

Learned how to use State as a geographical field and visualize sales distribution across different locations.

7. Dashboard Development

Learned how to arrange multiple visualizations into a single business dashboard.

8. Dashboard Formatting

Learned how to use titles, layouts, formatting, numerical display units, and consistent design to improve dashboard readability.

9. Business Analysis

Learned how to identify:

- Top-performing categories
- Top-performing sub-categories
- Strong and weak regions
- High-performing states
- Overall profitability

10. Data-Driven Decision Making

Gained practical knowledge of how Power BI can convert raw business data into meaningful visual insights that support managerial decision-making.